

Ixeia Sánchez Périz

DATA & AI AT SCALE

Zürich (Switzerland) • 📞 +41 78 230 9596 • ✉️ ixeiasp98@gmail.com
🌐 ixsanpe.github.io | www.linkedin.com/in/ixeia-s-p/ | 🎓 Scholar

SUMMARY

Senior Data & AI Engineer (MSc Data Science, ETH Zürich; **SOSP '25** co-author) working across **research and infrastructure**: 3+ years building TB-scale pipelines that ingest, clean, filter and curate **large-scale datasets** (image, audio, text), alongside research on distributed LLM training. Telecommunication engineering background in **signal processing**. Strong in **Python + PySpark**, statistics & EDA, near-duplicate detection & quality filtering.

RELEVANT WORK EXPERIENCE

SWISS INTERNATIONAL AIRLINES

Jun. 2024 – present | Zürich

Senior Data Engineer • promoted in 1 yr 9 mo

- Data processing at scale:** built and operated **TB-scale** ingestion, cleaning, filtering and transformation pipelines end to end with PySpark on Delta Lake/Parquet & Databricks Workflows, migrating legacy SQL — **12× speedup** (6h → 30m) and **50% lower** operational cost.
- Data quality & curation:** established managed quality, lineage and monitoring across 100+ data products. **Text preprocessing** with near-duplicate detection and heuristic quality filters; daily batch SLA to >99%.
- Iterative benchmarking:** built online + offline evaluation for a **production AI agent** (deterministic replay via interaction cassettes, benchmark suites, live monitoring) to measure how data and prompts affect model behaviour and iterate on processing strategies.
- Production tooling & velocity:** developed an internal Python SDK and CI/CD platform (Azure DevOps) used by 15 engineers — **36× faster deployment** (3h → 5min); drove team adoption of **AI coding agents**.
- Communication:** presented our Data Platform to Microsoft's Fabric product team (Redmond, US).

ETH ZÜRICH

Oct. 2023 – May 2024 | Zürich

LLM Research Engineer • MSc Thesis with Honours

- Publication:** **Sailor: Automating Distributed Training over Dynamic, Heterogeneous, and Geo-distributed Clusters**. F. Strati, Z. Zhang, G. Manos, I. Sánchez Périz, et al. *SOSP '25*. [doi:10.1145/3731569.3764839](https://doi.org/10.1145/3731569.3764839)
- Elastic fault tolerance:** implemented the fault-tolerance mechanism for a distributed LLM training system on PyTorch/DeepSpeed. Elastic recovery from node failures and spot preemptions via failure detection and reconfiguration with **2.87× higher throughput**. [Open-source](#).

MULTIMODAL DATA PROJECTS (video, audio & vision)

- Vision-language dataset** (fitness exercises) ([GitHub](#)): Curated and indexed image + text data into a vector database (Milvus) via joint embeddings for semantic retrieval; experiments tracked with MLflow.
- Audio preprocessing pipeline** (BirdCLEF field recordings) ([GitHub](#)): resampling, denoising, mel-spectrogram front-ends, augmentation with automated evaluation of **250+ configurations** measuring how preprocessing choices affect model performance.
- Streaming biomedical signal** (Corify Care): real-time ML classification over **64-channel** time-series on a medical device: filtering, resampling, artifact/dropout detection and windowed feature extraction under latency constraints.
- Video data stack benchmark** with **Ray Data/Lance vs. PySpark/Parquet**: decode → frame sampling → embedding → write over video data; measured throughput, on-disk size, random vs. sequential read latency and **indexed vector lookups** for shuffled training loads.
- Tree-crown segmentation on aerial imagery** ([GitHub](#)): image pipelines (cleaning, corrupted-image detection, cross-resolution downsampling) over COCO-annotated data; segmentation models (Detectron2, UNet/DeepLabv3+) with W&B sweeps.

SKILLS

LANGUAGES	Python, SQL Pandas, polars, NumPy Bash, Git
VIDEO & AUDIO	decoding, shot detection, frame sampling Digital Signal Processing spectrogram/mel features
DATA & CURATION	Statistics & EDA near-dedup, quality filtering, data mixes coverage & gap analysis lineage, versioning, observability
PROCESSING & STORAGE	PySpark, SparkSQL Ray Data Databricks Workflows Delta Lake, Lance, Parquet embeddings & vector search (Milvus)
ML & INFRA	PyTorch, DeepSpeed model eval & monitoring, MLflow Azure, AWS, GCP Docker, Kubernetes distributed systems

EDUCATION

ETH ZÜRICH

Sep. 2021 – Apr. 2024 | Zürich

MSc in Data Science (Computer Science Department)

POLYTECHNIC UNIVERSITY OF VALENCIA (UPV)

Sep. 2016 – Jul. 2021 | Valencia (Spain)

Double BS in Telecommunication Engineering & Business Administration and Management

- Ranked **1st** (avg. 9.1/10, 33 courses with honours) | Santander Scholarship, exchange programme in Singapore

HONORS & LANGUAGES

- Commencement Speaker, BSc graduation (2025) | Best Student Award by Social Council, 500+ students (2022) | Nova Top 111 talent in Spain under 25 (2022) | Outstanding Graduate, 1st in Class (2021).

LANGUAGES: English (C1) | German (B1/B2) | Spanish (Native)